



1. Motivation

- Standard recipe:** RLVR via GRPO [2, 4] runs after SFT and keeps the highest-scoring SFT checkpoint.
- The trap:** this can pick the **worst** checkpoint for RL. As SFT deepens, pass@1 rises but the GRPO pass@10 it reaches *falls* [1], so RL compute is spent on a doomed checkpoint.
- Why:** entropy collapse from SFT overtraining puts almost all mass on one continuation, so GRPO rollouts go homogeneous and the reward variance vanishes.
- Mechanism:** inside a GRPO group the advantage variance vanishes once pass@1 drops below $p^*(g)$ ($p^*(8)=0.083$), leaving most groups with no gradient:

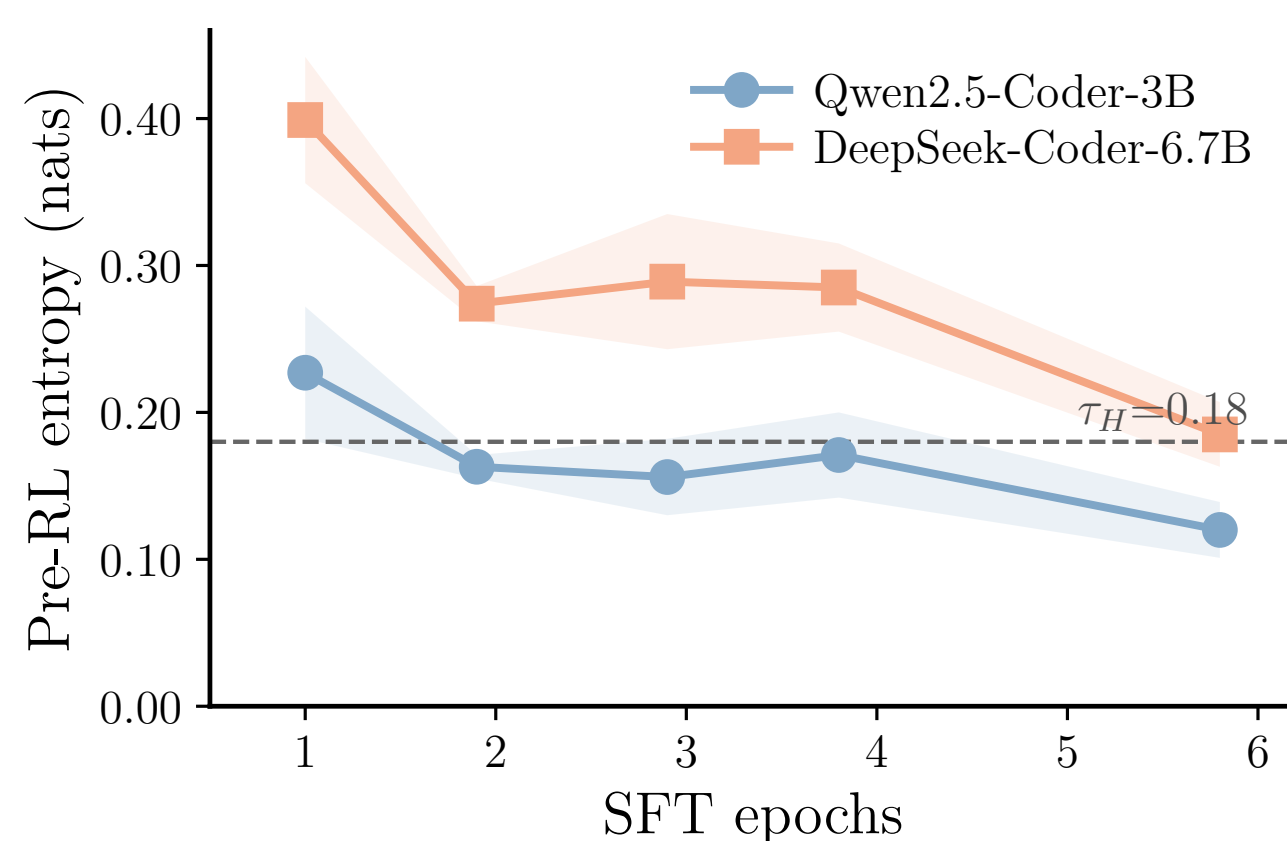
$$\mathbb{E}[\sigma_G^2] = \text{pass@1} (1 - \text{pass@1}) \cdot \frac{g-1}{g}$$

- Our fix:** an entropy screen that rejects doomed checkpoints at **zero RL cost**, validated on 3B and 6.7B code ladders.

2. Method: SFT-GRPO Pipeline

- Two SFT ladders:** Qwen2.5-Coder-3B-Base (primary) and DeepSeek-Coder-6.7B-Base (cross-model), same recipe, LoRA ($r=\alpha=128$).
- SFT data:** 5,000 KodCode-V1 problems with Gemini 2.5 Flash CoT traces, decontaminated against HumanEval+/MBPP.
- GRPO data:** held-out KodCode-V1 problems (test-suite verdict), pass count calibrated to [1, 14] ($\sim 1,100$ records).
- GRPO:** 400 steps, DAPO [4], $G=8$, $\beta=0$, $\varepsilon_{\text{high}}=0.28$, lr 1×10^{-6} , binary reward.
- Eval:** 40 frozen HumanEval+, $T=1.0$, 3 seeds. pass@10 every 50 steps ($n=20$), deep pass@10 ($n=128$) for selection. **Entropy:** mean next-token entropy ($-\sum_v p_v \log p_v$ nats).

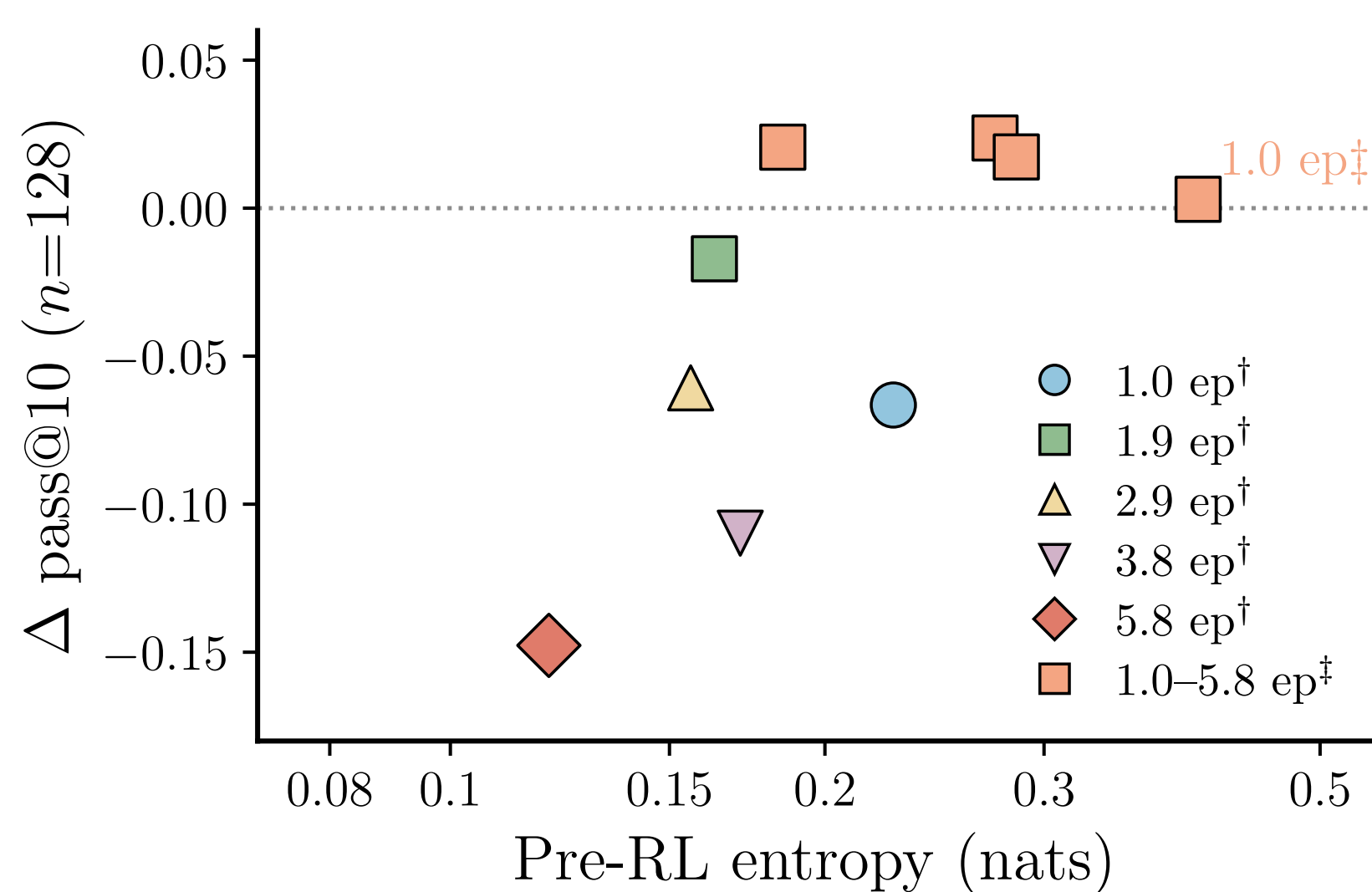
3. Pre-RL Entropy Ladder



- Screens pre-RL:** Qwen's pass@1 *rises* with depth while entropy *falls*, so the top scorer is the **worst** pick.

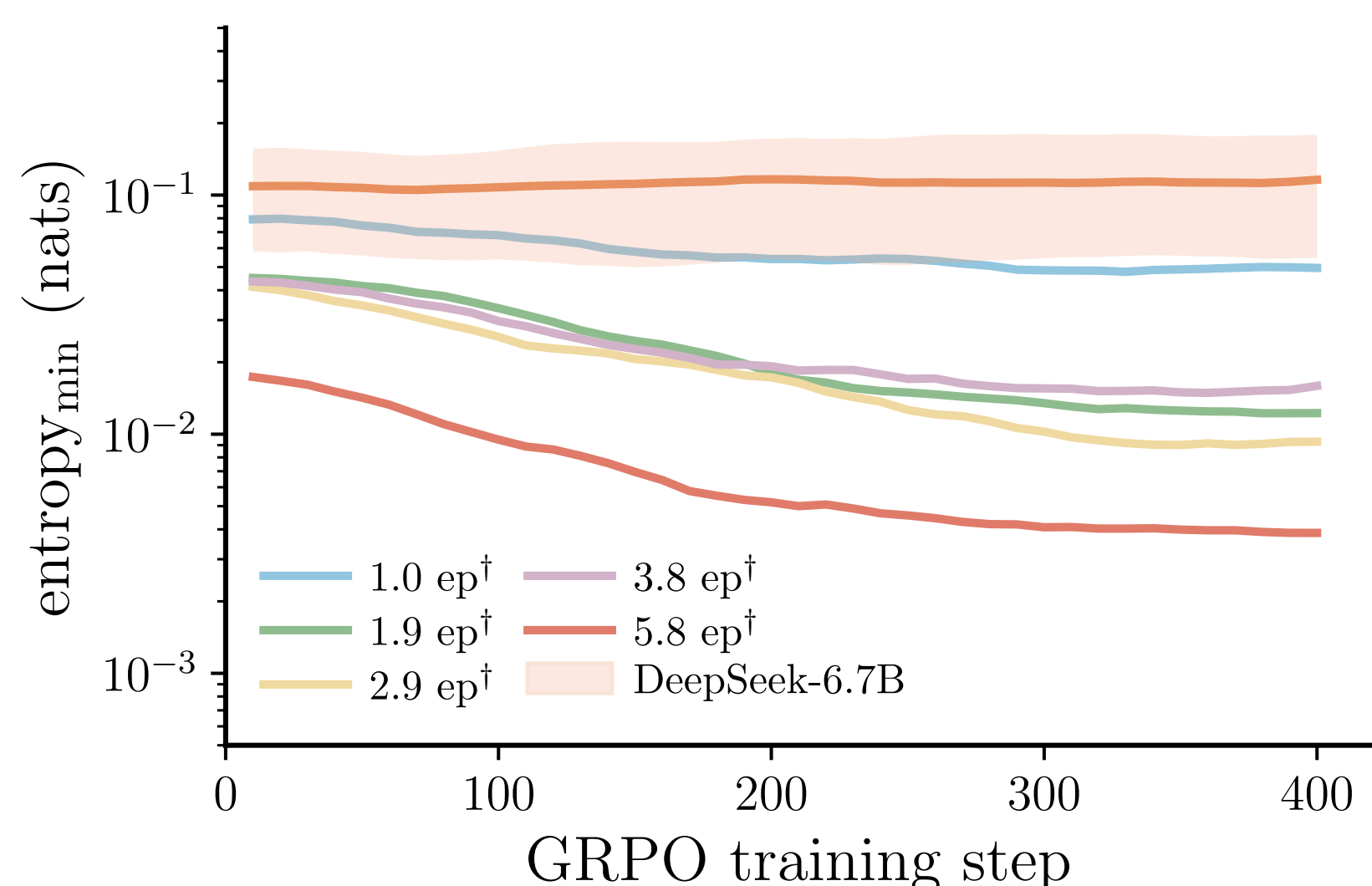
- Verdict:** Qwen bottoms at 0.12 nats; DeepSeek stays ≥ 0.185 . τ_H predicts Qwen collapses, DeepSeek does not.

4. $\Delta \text{pass@10}$ vs. Pre-RL Entropy



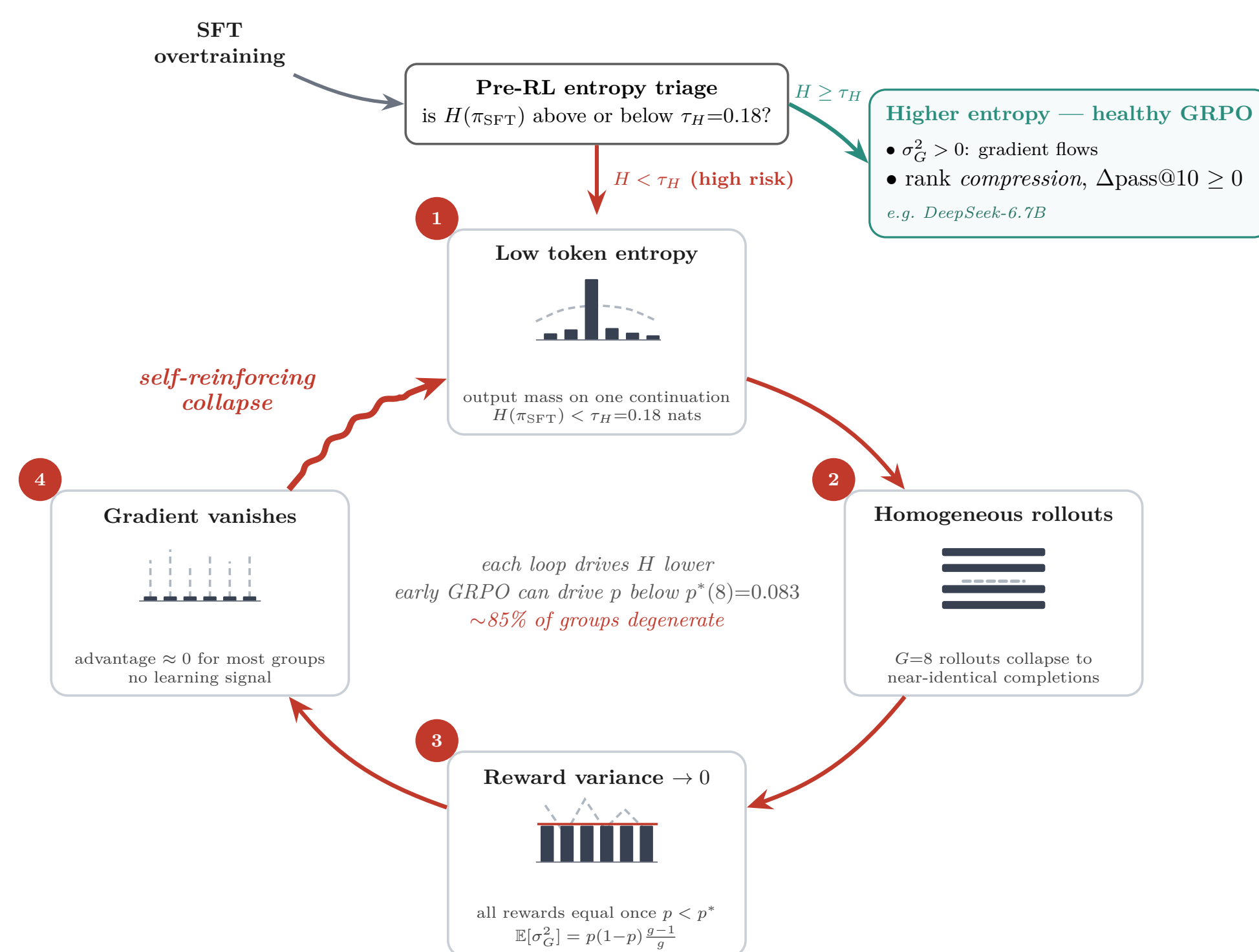
- Lower pre-RL entropy gives **inversion** (Qwen, $\Delta < 0$); higher entropy gives compression (DeepSeek, $\Delta \geq 0$). Dashed: the threshold $p^*(8)=0.083$ where the gradient vanishes.

5. Entropy Collapse During GRPO



- Entropy keeps falling *during* RL: the deepest Qwen loses 79%, ending 12 $\times$ below the shallowest, while DeepSeek stays flat above $p^*(8)$.

6. Mechanism: The Entropy-Collapse Spiral



7. Two-Stage Diagnostic

- Stage 1: pre-RL entropy screen** (no RL compute). Reject a checkpoint if

$$H(\pi_{\text{SFT}}) < \tau_H = 0.18 \text{ nats}$$

rejecting a doomed checkpoint saves **100%** of its RL budget.

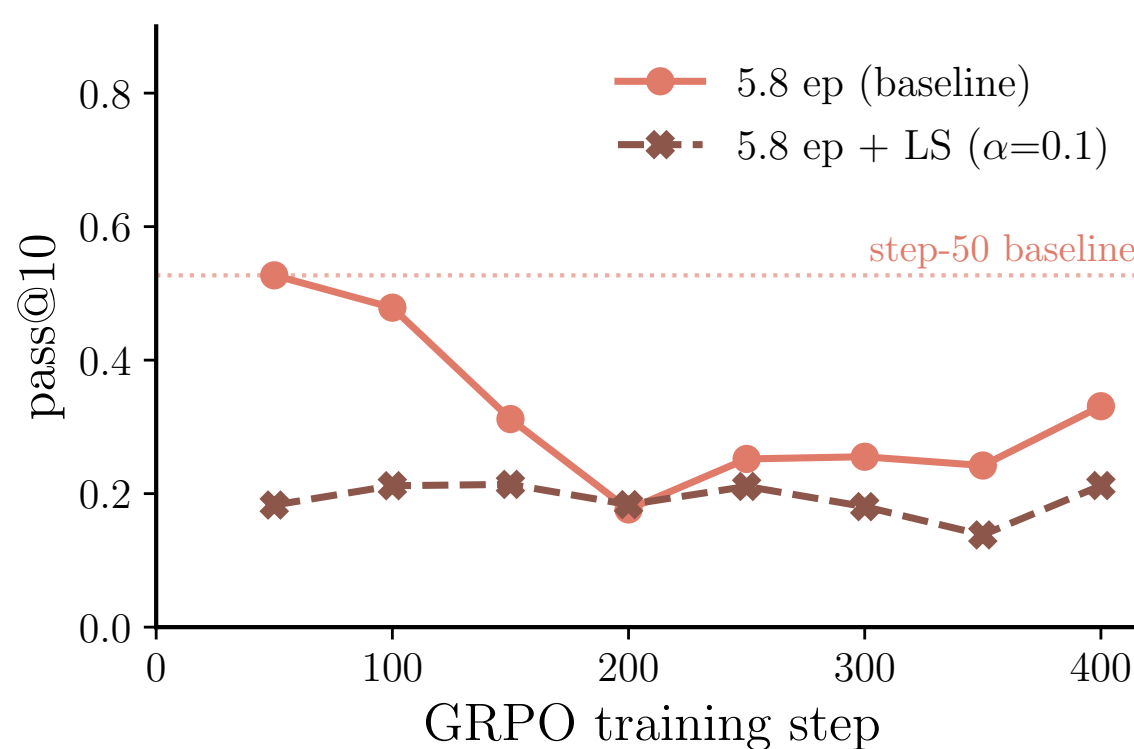
- Stage 2: step-150 monitor.** Stop the run if the relative entropy drop

$$\frac{\Delta H(10 \rightarrow 150)}{H(10)} > \tau_2 = 0.50$$

saving **62.5%** of the run.

- Selection gain:** the diagnostic picks the 1.9-epoch checkpoint over the standard “deepest” (5.8-epoch) rule, lifting deep-eval pass@10 **0.553** $\rightarrow$ **0.643** (+0.090, $n=128$).

8. Interventions Do Not Rescue Collapse



- KL penalty** ($\beta=0.01$): rank inversion preserved; structural damage, not a KL artefact.
- Label smoothing** ($\alpha=0.1$): *highest* pre-RL pass@1 (0.600) yet *lowest* peak pass@10 (0.0214): the **restoration paradox**.
- Together they rule out hyper-parameter and RL-stage explanations.

9. Key Results and Takeaways

- Rank inversion (Qwen):** peak GRPO pass@10 falls **0.806** $\rightarrow$ **0.481** with SFT depth, peaking earlier (step 250 $\rightarrow$ 50); also seen in math reasoning [3].
- DeepSeek compresses, not inverts:** the 6.7B ladder bunches, post-RL rank = pre-RL ($\rho=+1.00$).
- Entropy beats pass@1:** Spearman ρ with peak pass@10 is +0.69 for pre-RL entropy vs. -0.75 for pass@1 ($n=15$, $p<0.01$); higher pass@1 predicts worse RL.
- Threshold mechanism:** by step 200 Qwen's pass@1 = $0.020 < p^*(8)=0.083$, so $\sim 85\%$ of groups give zero gradient.
- More rollouts don't help:** doubling g only halves $p^*(g)$; a collapsed policy stays collapsed.

Practitioner Takeaway: Measure mean token entropy before spending RL compute. If pre-RL entropy $< \tau_H \approx 0.18$ nats (or $\text{pass@1} < p^*(g)$), expect zero-gradient collapse and ladder inversion. Optimising pass@1 directly (label smoothing) can *worsen* RL: the restoration paradox.

10. Conclusion

- Entropy collapse** plus the formal threshold $p^*(g)$ jointly predict whether GRPO *inverts* (Qwen) or *compresses* (DeepSeek) the SFT ladder; pass@ k alone misses this.
- Pre-RL entropy** is a better RL-readiness signal than pass@ k , and the diagnostic acts before or early in RL.
- Future work:** test the $p^*(g)$ diagnostic on math/reasoning and non-binary rewards, larger g , and generalisation-loss predictors.

References

- [1] Kang, F., Kuchnik, M., Padthe, K., Vlastelica, M., Jia, R., Wu, C.-J., and Ardalani, N. Quagmires in SFT-RL post-training: When high SFT scores mislead and what to use instead. *arXiv preprint arXiv:2510.01624*, 2025.
- [2] Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [3] Zhang, D., Xu, Y., Wang, H., Chen, Q., and Peng, H. Good SFT optimizes for SFT, better SFT prepares for reinforcement learning. *arXiv preprint arXiv:2602.01058*, 2026.
- [4] Yu, Q. et al. DAPO: An open-source LLM reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.